

# Course Map and the Semester Project

DATA 443/643 · Advanced Concepts in Large Language Models

Week 1

American University

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# Lecture plan

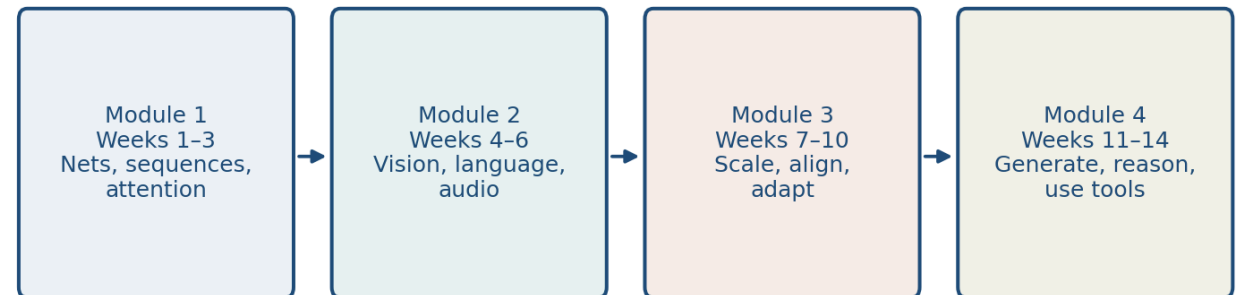
- |   |                                                      |       |
|---|------------------------------------------------------|-------|
| 1 | Where this course sits, and why it is self-contained | 4 min |
| 2 | An LLM is a stack; Week 1 is the base                | 6 min |
| 3 | What pretraining actually minimizes                  | 6 min |
| 4 | The 140 GB weights file                              | 4 min |
| 5 | Turn a method into a project question                | 6 min |

Key goal: leave with a stack picture, a next-token loss, and one measurable project sentence.

# A self-contained LLM course

- Listed prereq: DATA 427/627.
- DATA 641 and 642 are not required.
- Overlap is intentional so more people can finish a project.

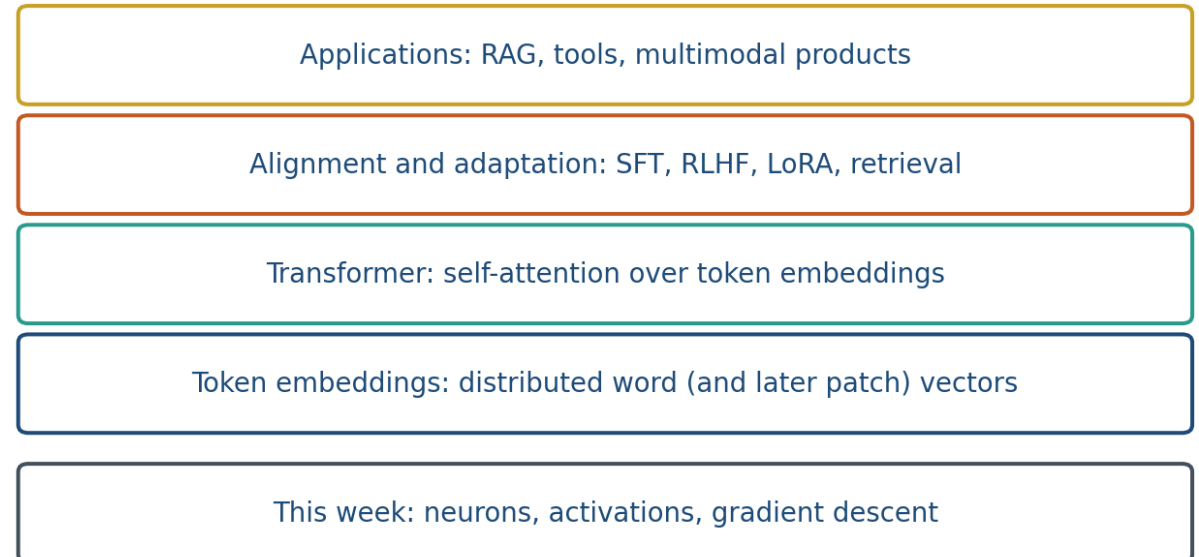
Where Week 1 sits: foundations before transformers



# A language model is a stack

- Chat products sit at the top.
- The weights still came from next-token training.
- Name the layer you will change: data, prompt, adapter, retriever, or eval.

A language model is a stack. Week 1 is the bottom two layers.



# Pretraining minimizes next-token NLL

$$P(w_1, \dots, w_T) = \prod_{t=1}^T P(w_t \mid w_{1:t-1})$$

Each factor is a softmax over the vocabulary.

The training loss is the negative log of those probabilities.

If the model assigns 0.70, NLL is about 0.36. If 0.10, NLL is about 2.30.

# Llama 2 70B is a 140 GB file

$$70 \times 10^9 \times 2 \text{ bytes} = 140 \text{ GB}$$

Each parameter stored as float16 (2 bytes).

Parameters file vs run file: running needs extra memory.

Karpathy: an LLM is weights plus a little code.

# The project starts with a question

## Not a project

Use LoRA on news.

## A project

Does a LoRA adapter on local news reduce entity hallucination vs. the base model, measured by exact-match on a 100-item holdout?

# Today's two-hour meeting

## 1.1 ~26 min

Stack, NLL, 140 GB, project sentence. Karpathy 0:00–8:00.

## 1.2–1.4 ~52 min

Neuron / XOR, then GD + one chain-rule, then skip-gram softmax + bias.

## Lab 1 ~32 min

XOR vs linear, then the constructed 2-D embedding probe.

Pedagogy drawn from Stanford CS224N (plan + one formula), CS231N (footer, one idea), MIT 6.S191 (diagram + numbers). Original slides; those courses are not copied.